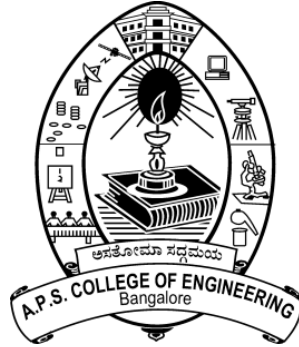


APS COLLEGE OF ENGINEERING

DEPARTMENT OF ECE



Electronic Principles and Circuits - BEC303

LABORATORY MANUAL

EVEN SEMESTER – 2023-2024

Lab Incharge : Prof. Purva Kamat Mhamai

Instructor : Mrs Mamatha Dixit

Name: _____

USN : _____

VISVESVARAYA TECHNOLOGICAL UNIVERSITY, BELAGAVI

B.E: Electronics & Communication Engineering / B.E: Electronics & Telecommunication Engineering

NEP, Outcome Based Education (OBE) and Choice Based Credit System (CBCS)

(Effective from the academic year 2023 – 24) -III Semester

Electronic Principles and Circuits - BEC303

Experiments using Pspice software

1. Design and Test
 - (i) Bridge Rectifier with Capacitor Input Filter.
 - (ii) Zener voltage regulator.
2. Design and Test
 - 1) Biased Clippers – a) Positive, b) Negative, c) Positive-Negative
 - 2) Positive and Negative Clampers with and without Reference.
3. Plot the transfer and drain characteristics of a JFET and calculate its drain resistance, mutual conductance and amplification factor.
4. Plot the transfer and drain characteristics of n-channel MOSFET and calculate its parameters drain resistance, mutual conductance and amplification factor.
5. Design and test (i) Emitter Follower (ii) Darlington Connection.
6. Design and plot the frequency response of Common Source JFET/MOSFET amplifier.
7. Test the Opamp Comparator with zero and non zero reference and obtain the Hysteresis curve.
8. Design and test Full wave Controlled rectifier using RC triggering circuit.
9. Design and test Precision Half wave and full wave rectifiers using Opamp.
10. Design and test RC phase shift oscillator.

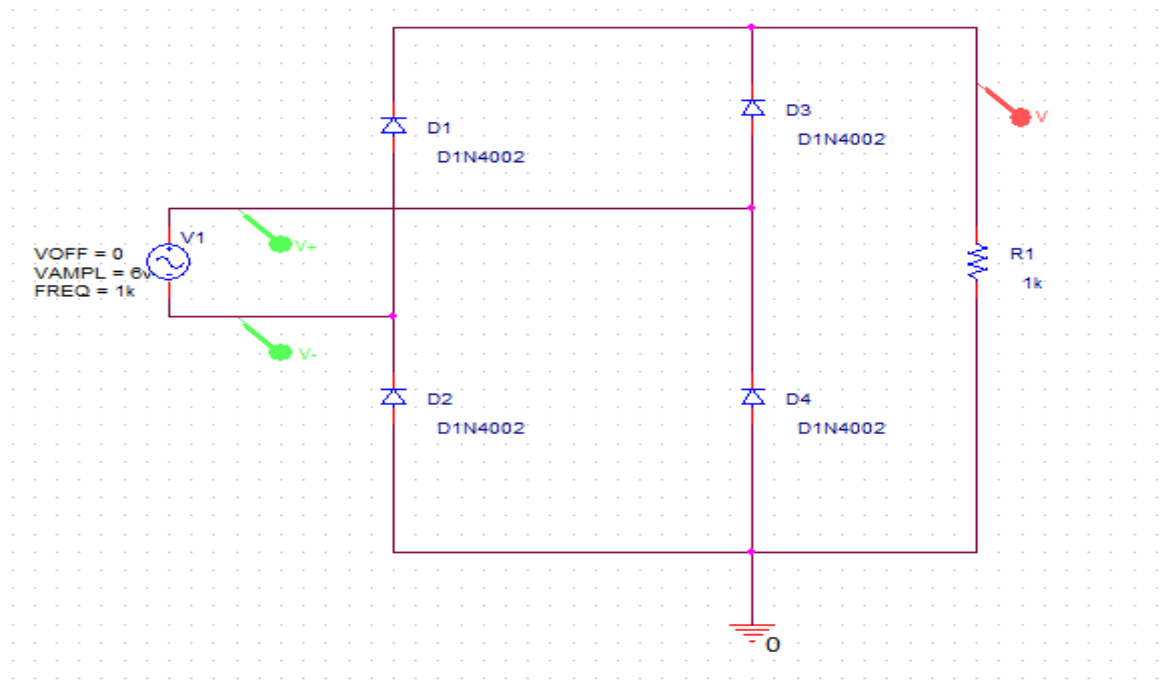
PROCEDURE FOR PSPICE:

1. Go to START → PROGRAMS→PSPICE STUDENT→CAPTURE.
2. Go to FILE→NEW→PROJECT.
3. Give a name to the Project and also select “ ANALOG AND A/D”.
4. Select the path C: ORCADDEMO/PSPICE under Browse
5. Draw the circuit as shown in the circuit diagram by using the Drag and Drop Method and selecting the required component.
6. Save it and Go to PSPICE→NEW SIMULATION PROFILE.
7. Give a name and click OK.
8. Go to PSPICE→EDIT SIMULATION PROFILE.
9. Go to ANALYSIS→TIME DOMAIN(TRANSIENTS)
10. General Settings →Run to Time:100ms
11. Start saving Data → 0 seconds
12. Save and Go to PSPICE→RUN.
13. The Output Window appears.
14. Go to ADDTRACE→ Select the appropriate parameters to be displayed.
15. Observe and save the output waveform.

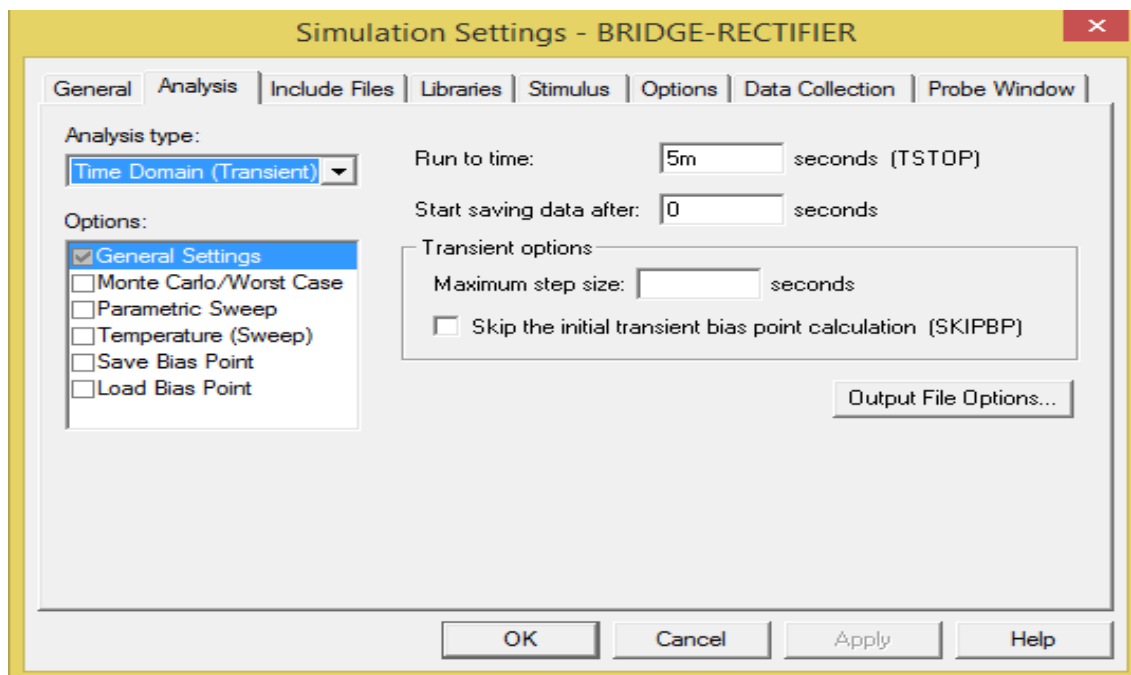
EXPERIMENT No: 1

AIM: Design and Test:

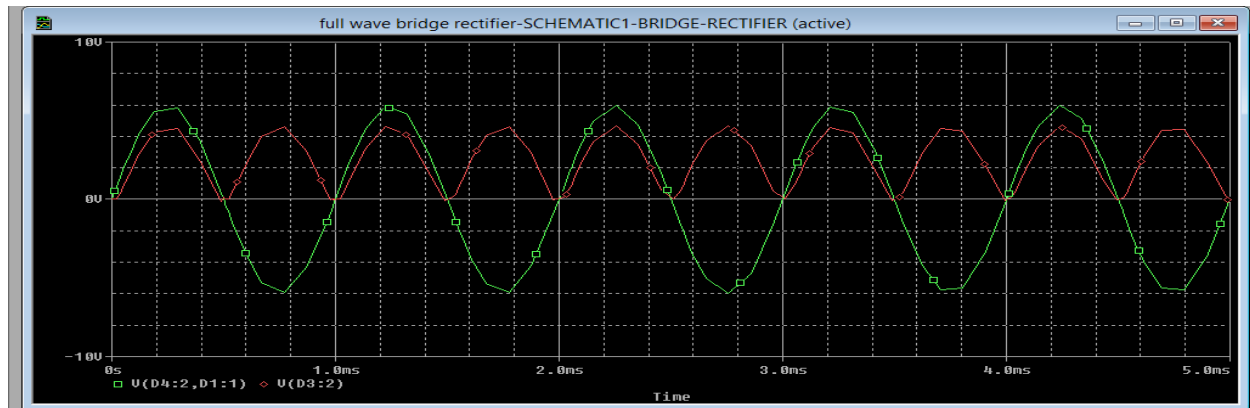
1 a). (i) Bridge Rectifier with Capacitor Input Filter:



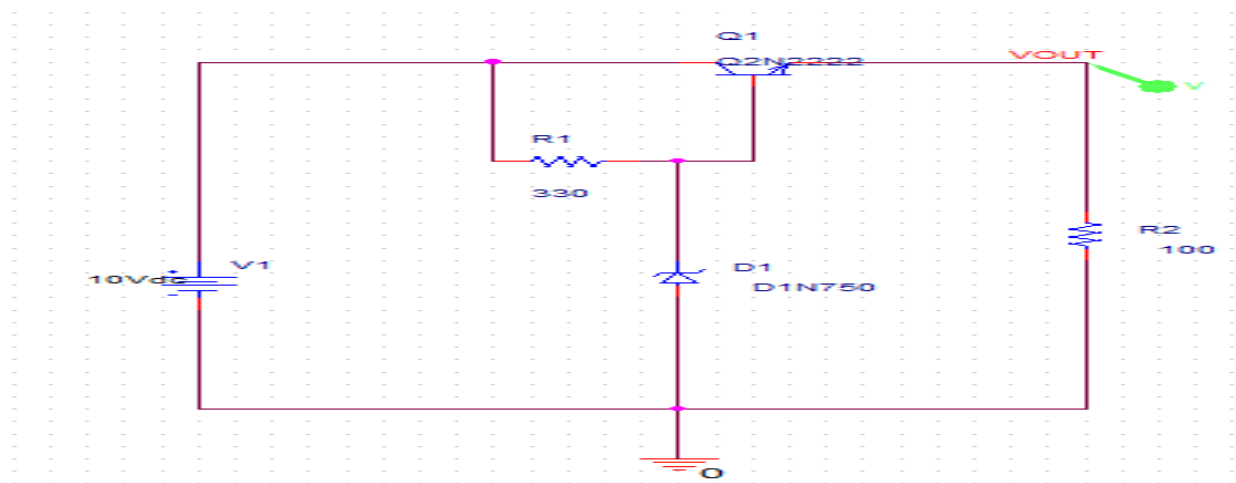
SIMULATION PROFILE SETTING :



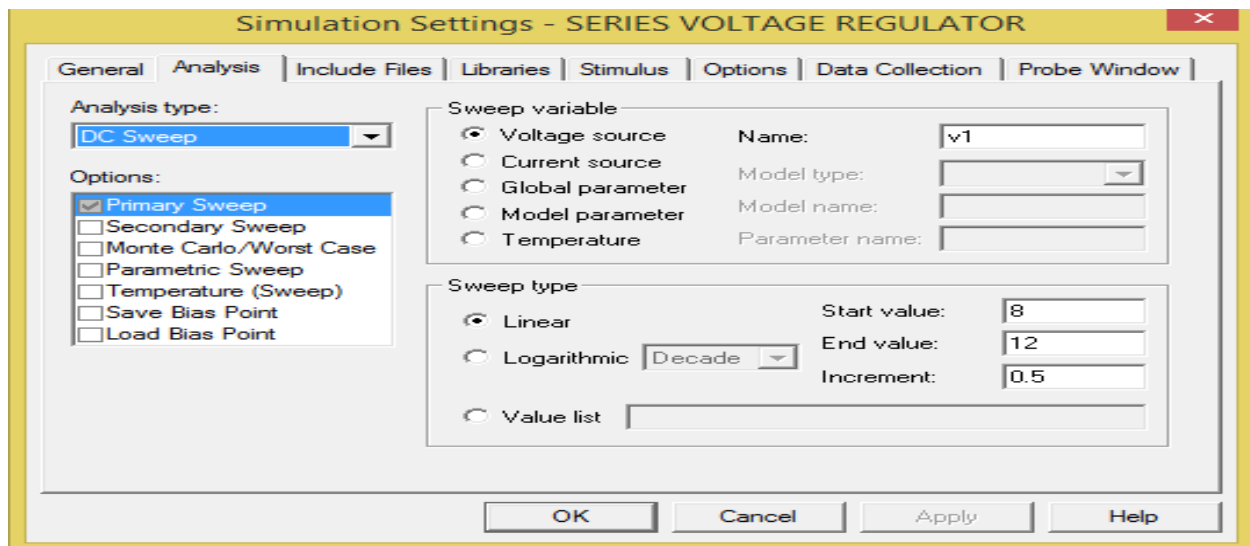
OUT PUT WAVEFORM :



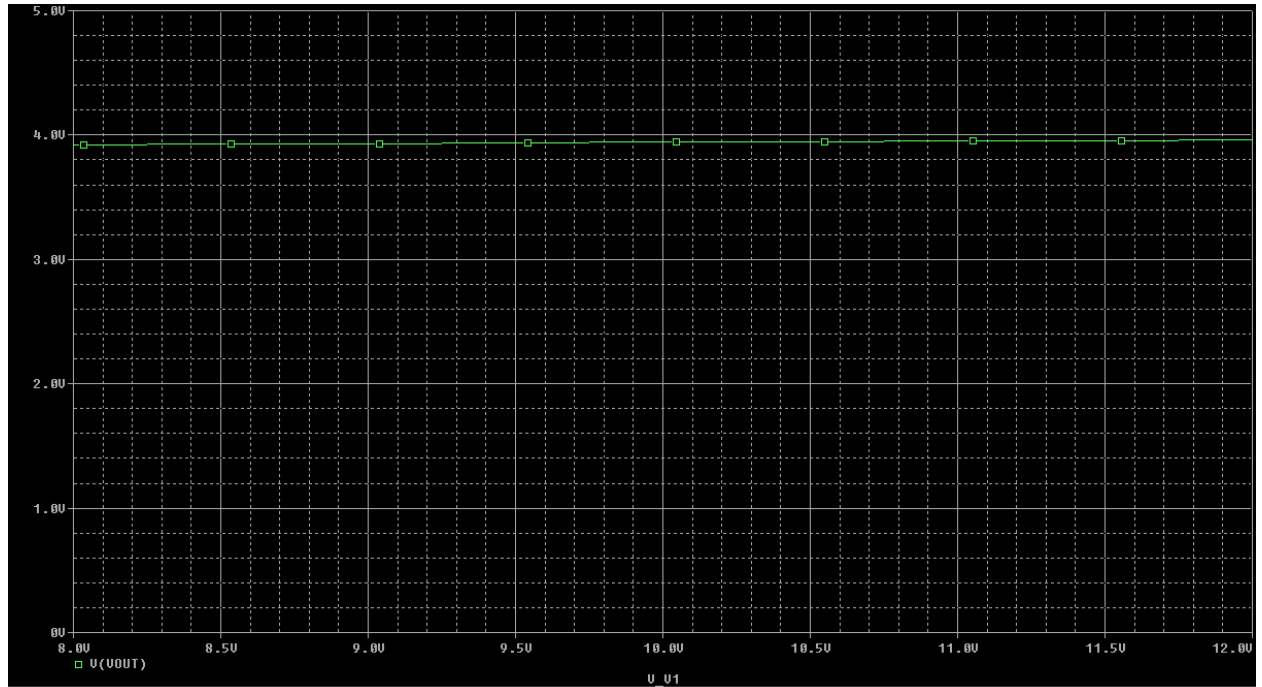
(ii) Zener voltage regulator :



SIMULATION PROFILE SETTING :



OUT PUT WAVEFORM :

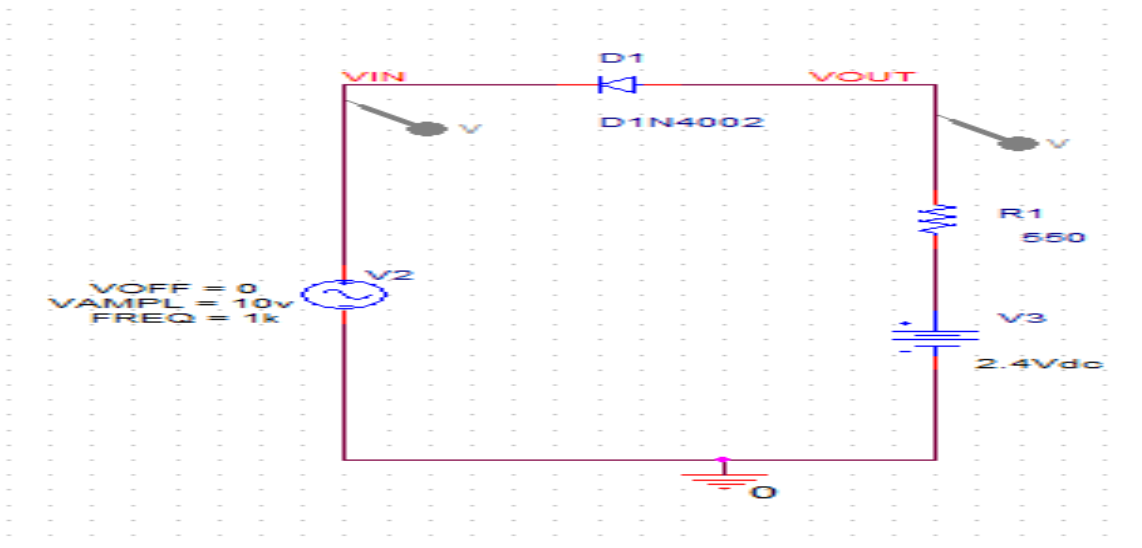


EXPERIMENT No: 2

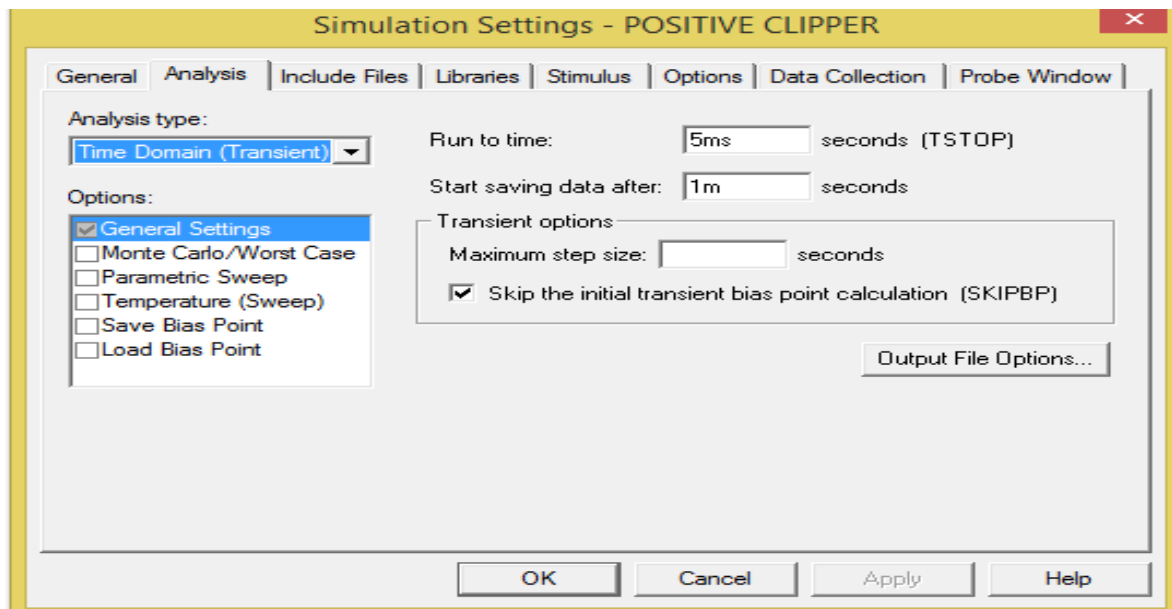
AIM : Design and Test

Biased Clippers – a) Positive b) Negative c) Positive-Negative

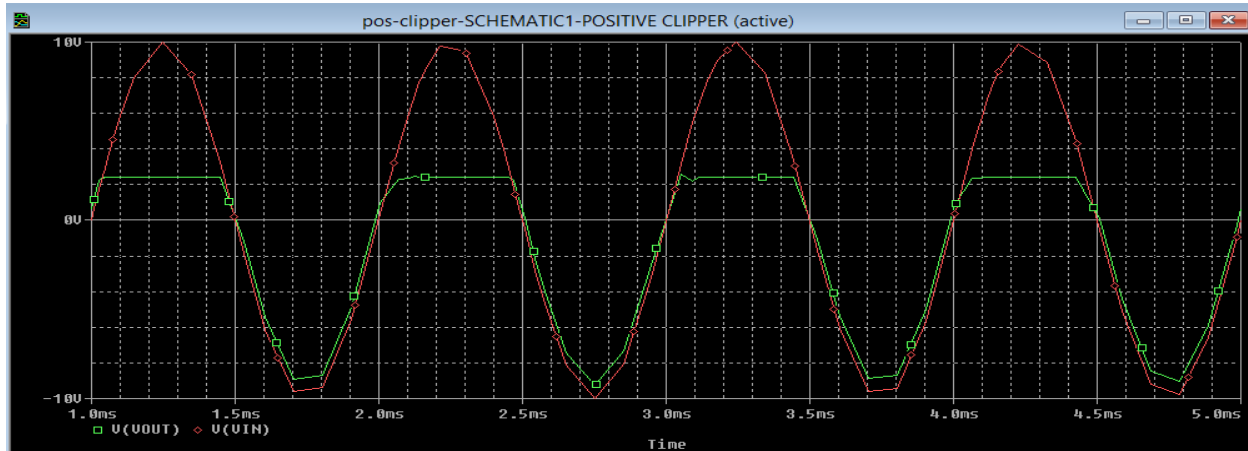
a) Positive Clipper :



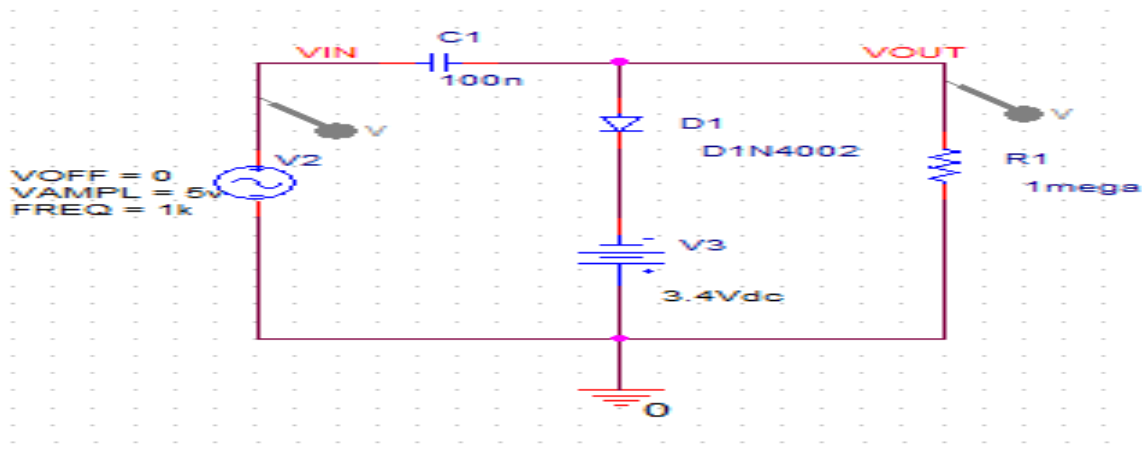
SIMULATION PROFILE SETTING :



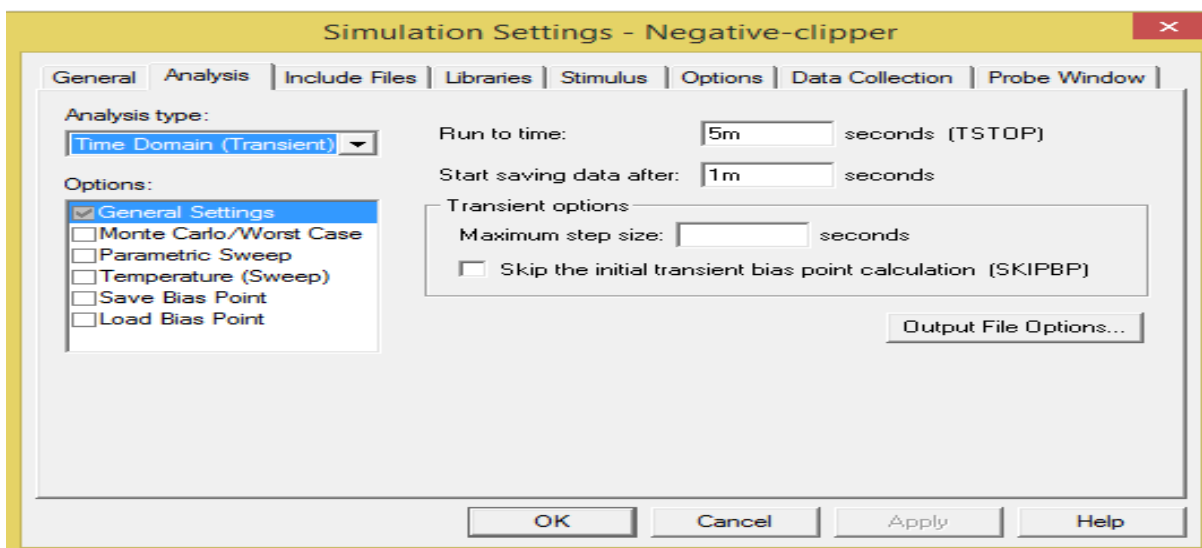
OUT PUT WAVEFORM :



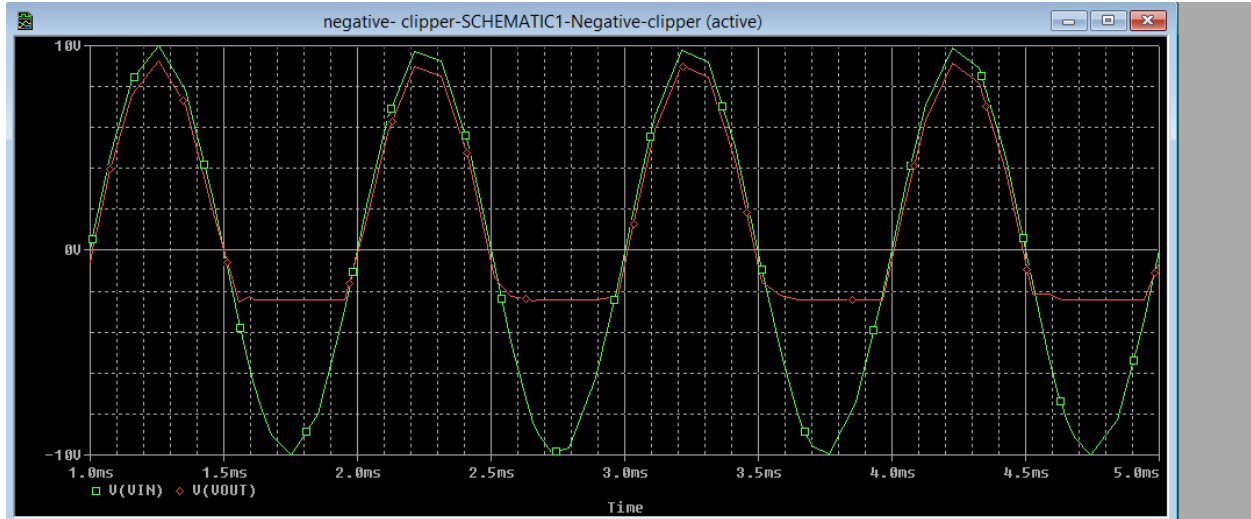
, b) Negative Clipper



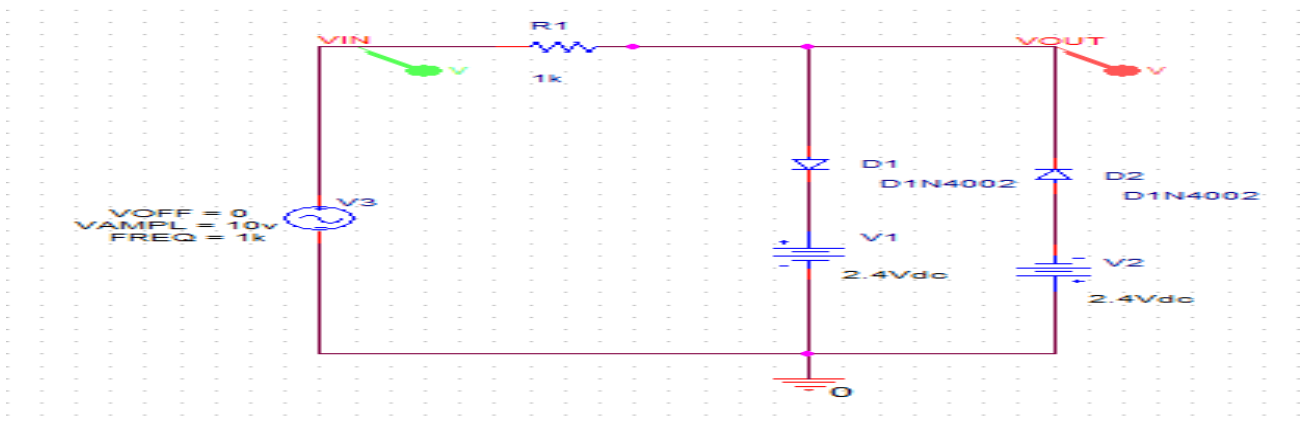
SIMULATION PROFILE SETTING :



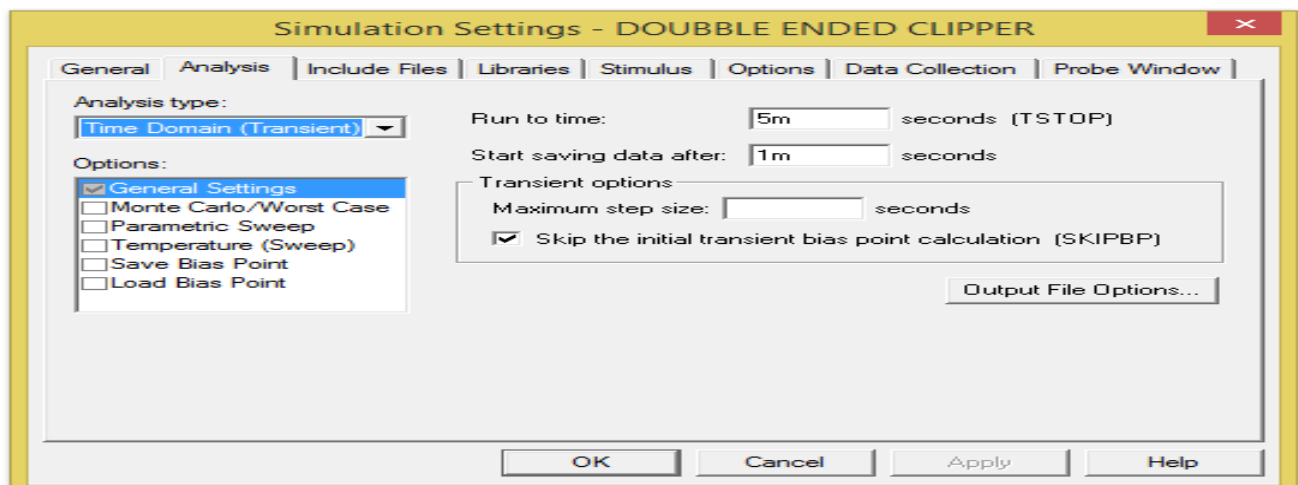
OUT PUT WAVEFORM :



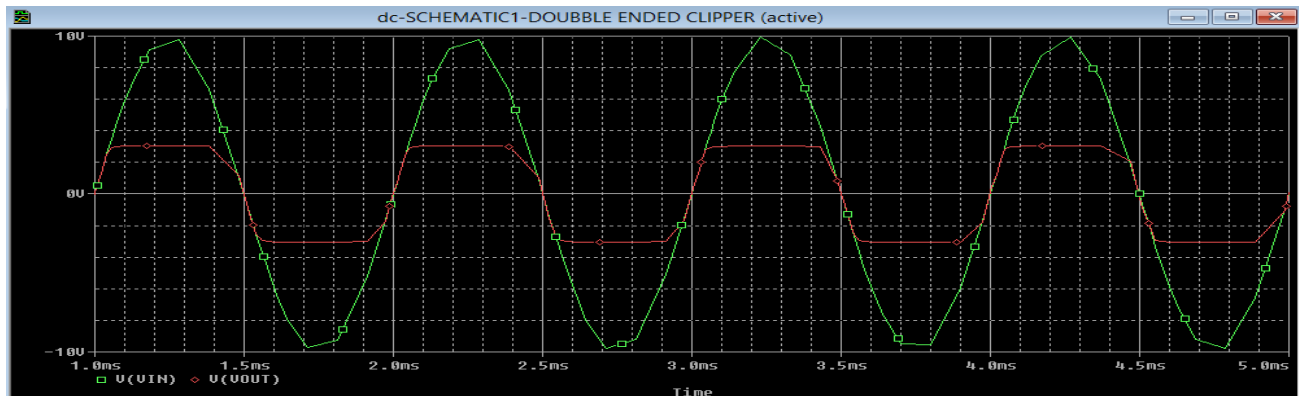
c) Positive-Negative



SIMULATION PROFILE SETTING :

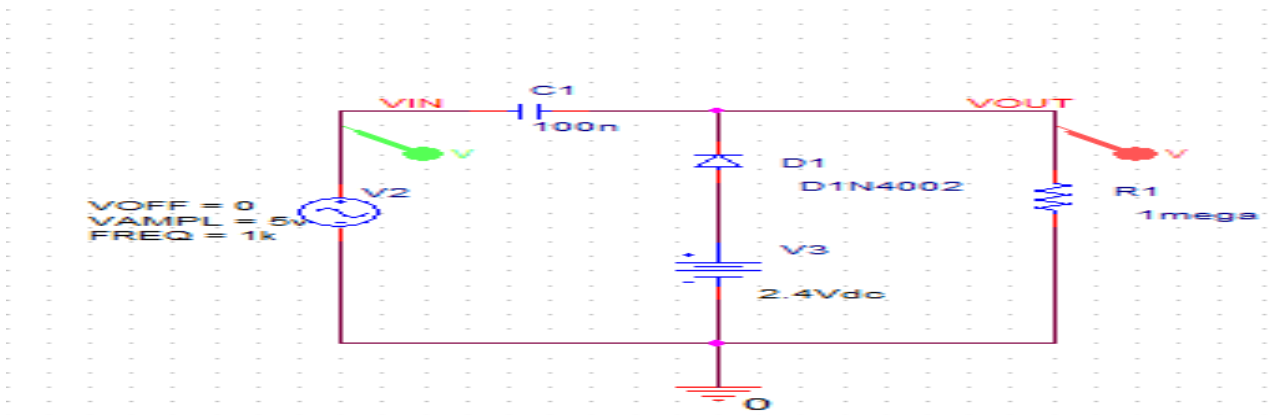


OUT PUT WAVEFORM :

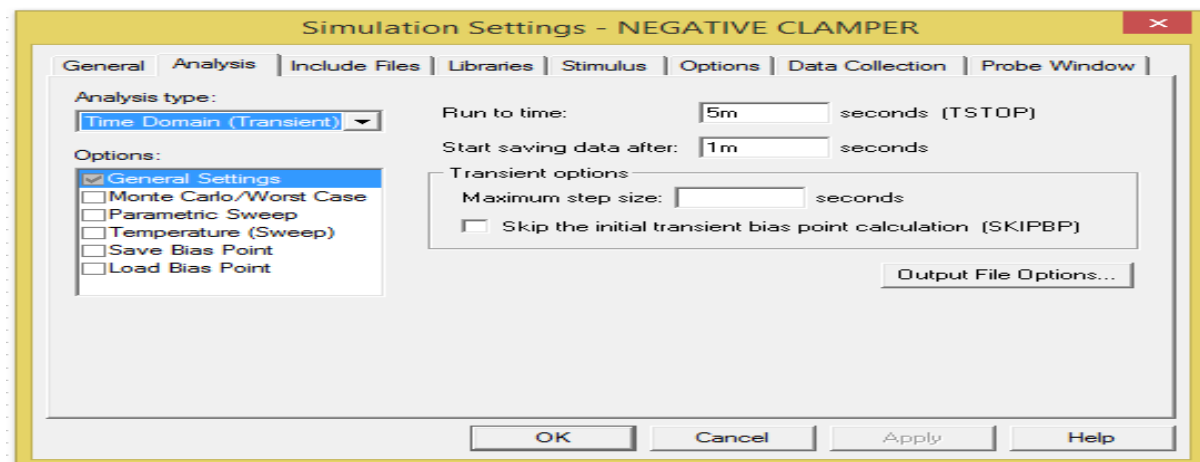


Positive and Negative Clampers with and without Reference.

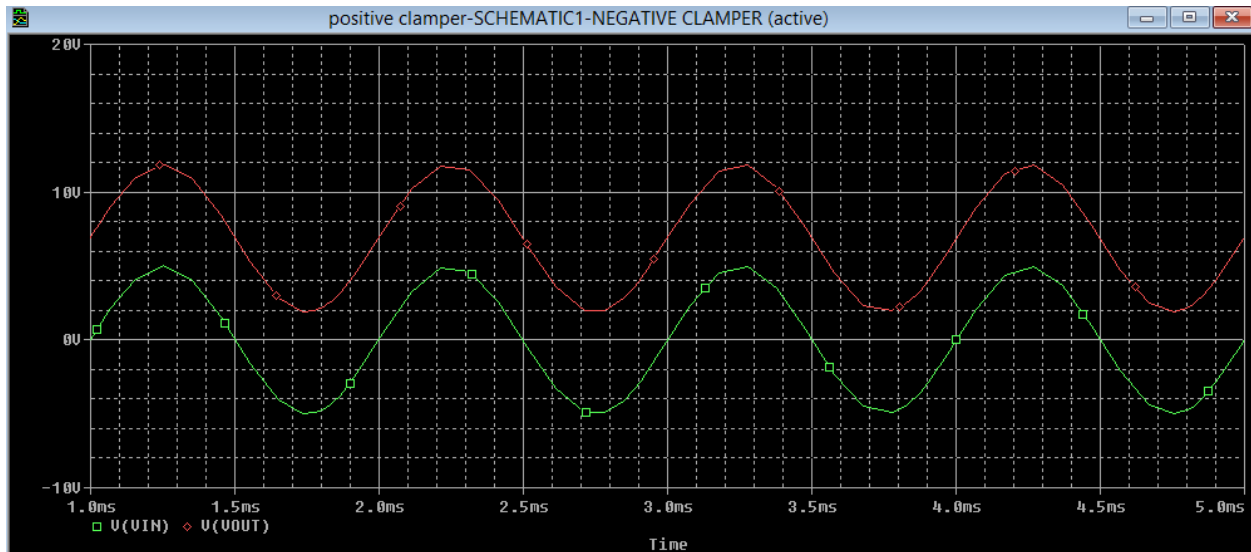
ii) Positive Clamper :



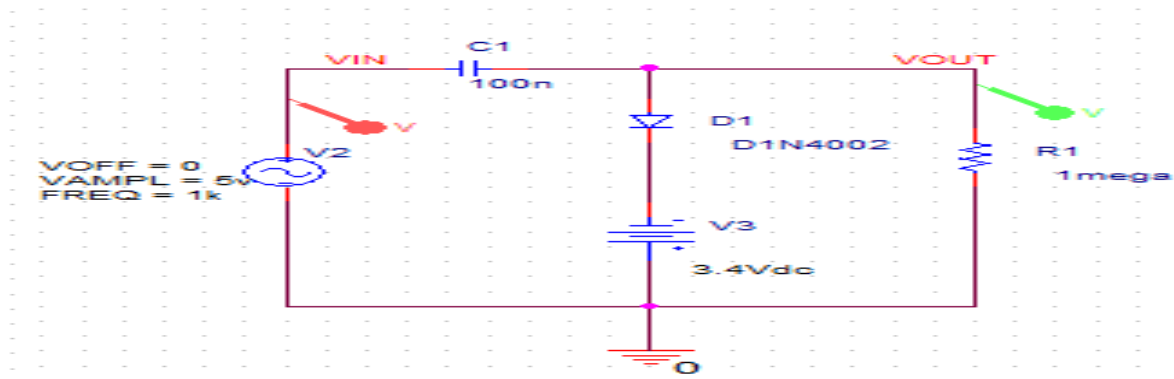
SIMULATION PROFILE SETTING :



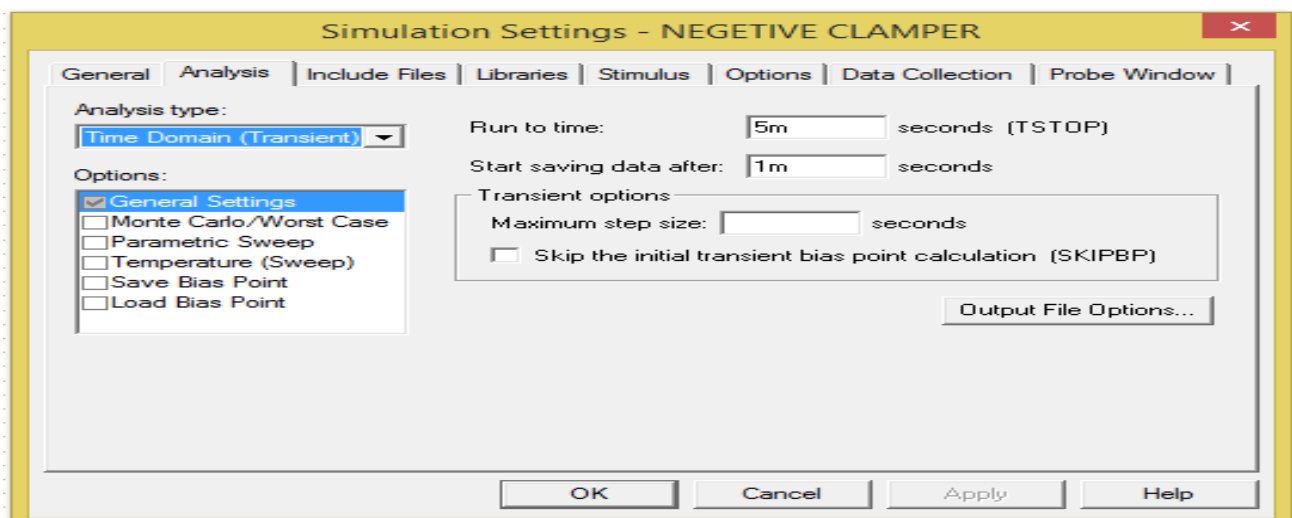
OUTPUT WAVEFORM :



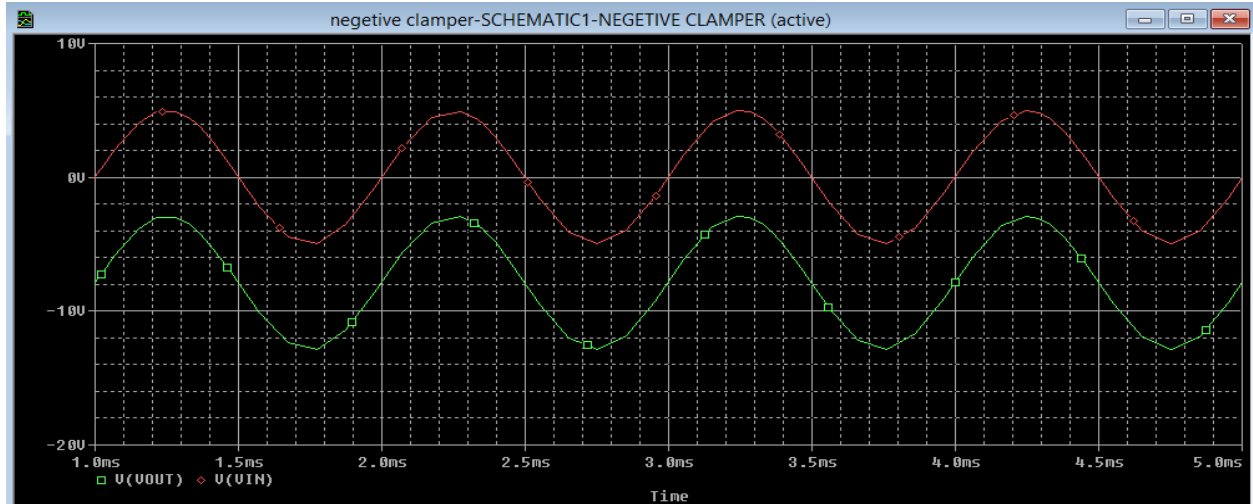
Negative Clampers :



SIMULATION PROFILE SETTING :



OUTPUT WAVEFORM :

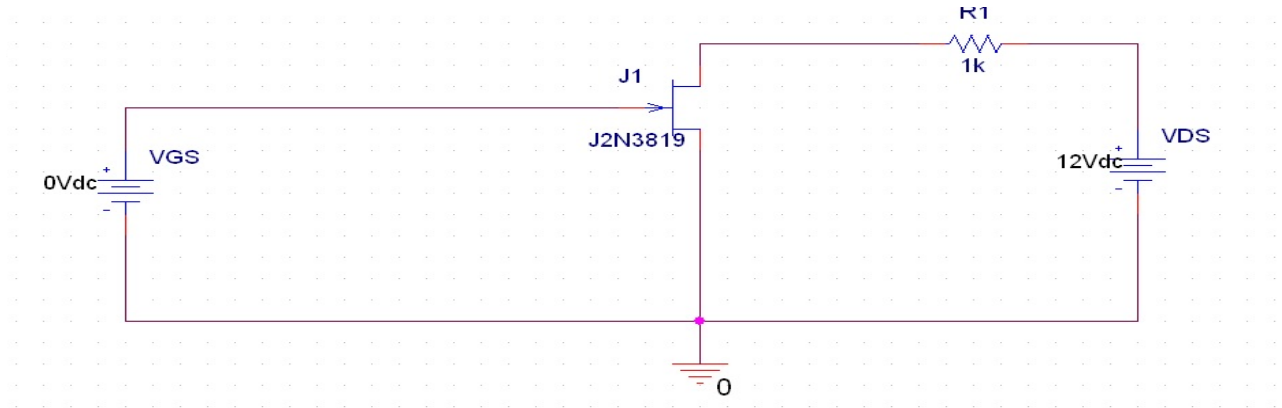


EXPERIMENT NO: 3

Transfer and drain characteristics of a JFET

AIM: To simulate and plot the Drain and Transfer characteristics of a JFET using pspice

CIRCUIT DIAGRAM:



PROCEDURE:

1. Go to START → PROGRAMS→PSPICE STUDENT→CAPTURE.
2. Go to FILE→NEW→PROJECT.
3. Give a name to the Project and also select “ ANALOG AND A/D”.
4. Select the path C: ORCADDEMO/PSPICE under Browse
5. Draw the circuit as shown in the circuit diagram by using the Drag and Drop Method and selecting the required component.
6. Save it and Go to PSPICE→NEW SIMULATION PROFILE.
7. Give a name and click OK.
8. Go to PSPICE→EDIT SIMULATION PROFILE.
9. Go to ANALYSIS→DC SWEEP
10. For the Transfer Characteristics:
Select PRIMARY SWEEP: VOLTAGE SOURCE: Vgs
Start Value: 0V
End Value: -3V
Increment :0.1V
Also Set the Value of Vds in the circuit diagram to 12 V.
11. Save and Go to PSPICE→RUN.
12. The Output Window appears.
13. Go to ADDTRACE→ Select the appropriate parameters to be displayed.
14. Observe and save the output waveform.

15. For the Drain Characteristics:

Select PRIMARY SWEEP: VOLTAGE SOURCE: V_{ds}

Start Value: 0V

End Value: 10V

Increment :0.1V

Select SECONDARY SWEEP: VOLTAGE SOURCE: V_{gs}

Start Value: -4V

End Value: 0V

Increment :1V

16. Save and Go to PSPICE→RUN.

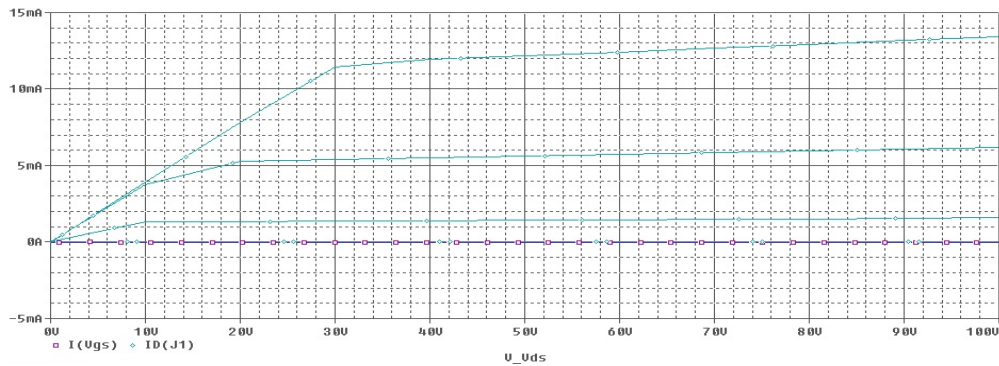
17. The Output Window appears.

18. Go to ADDTRACE→ Select the appropriate parameters to be displayed.

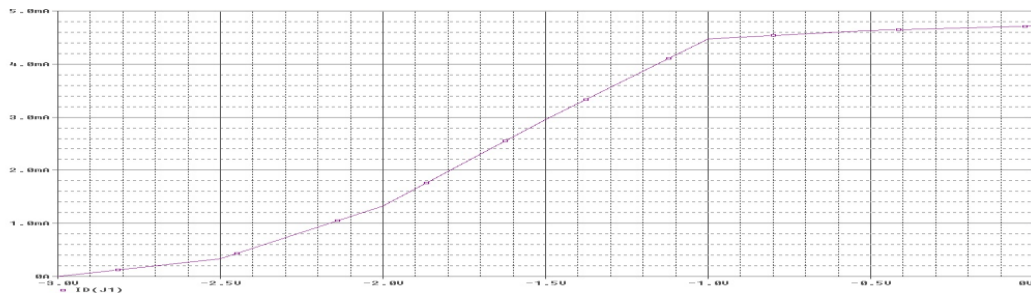
19. Observe and save the output waveform.

WAVEFORMS:

Drain Characteristics



Transfer Characteristics



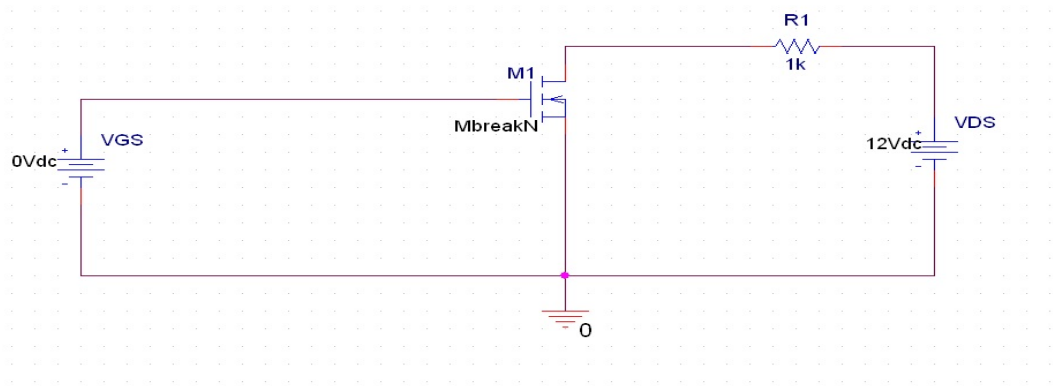
RESULT: The Transfer and Drain Characteristics of JFET were Plotted.

EXPERIMENT NO: 4

Transfer and drain characteristics of a MOSFET

AIM: To simulate and plot the Drain and Transfer characteristics of a MOSFET using pspice

CIRCUIT DIAGRAM:



PROCEDURE:

1. Go to START → PROGRAMS → PSPICE STUDENT → CAPTURE.
2. Go to FILE → NEW → PROJECT.
3. Give a name to the Project and also select “ANALOG AND A/D”.
4. Select the path C: ORCADDEMO/PSPICE under Browse
5. Draw the circuit as shown in the circuit diagram by using the Drag and Drop Method and selecting the required component.
6. Save it and Go to PSPICE → NEW SIMULATION PROFILE.
7. Give a name and click OK.
8. Go to PSPICE → EDIT SIMULATION PROFILE.
9. Go to ANALYSIS → DC SWEEP
10. For the Transfer Characteristics:
Select PRIMARY SWEEP: VOLTAGE SOURCE: Vgs
Start Value: 0V
End Value: 10V
Increment :0.1V
Also Set the Value of Vds in the circuit diagram to 12 V.
11. Save and Go to PSPICE → RUN.
12. The Output Window appears.
13. Go to ADDTRACE → Select the appropriate parameters to be displayed.
14. Observe and save the output waveform.

15. For the Drain Characteristics:

Select PRIMARY SWEEP: VOLTAGE SOURCE: V_{ds}

Start Value: 0V

End Value: 10V

Increment :1V

Select SECONDARY SWEEP: VOLTAGE SOURCE: V_{gs}

Start Value: 0V

End Value: 3V

Increment :1V

16. Save and Go to PSPICE→RUN.

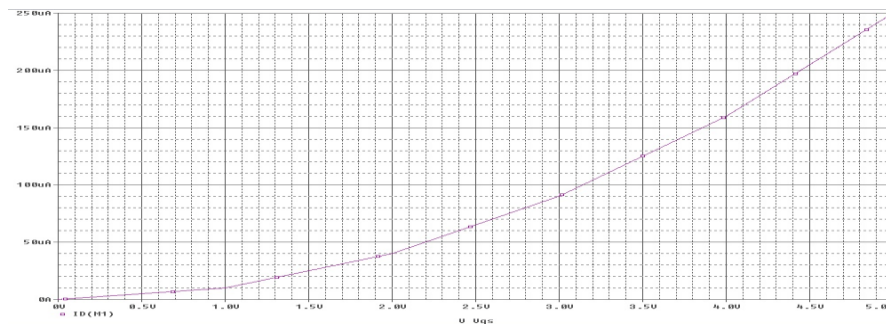
17. The Output Window appears.

18. Go to ADDTRACE→ Select the appropriate parameters to be displayed.

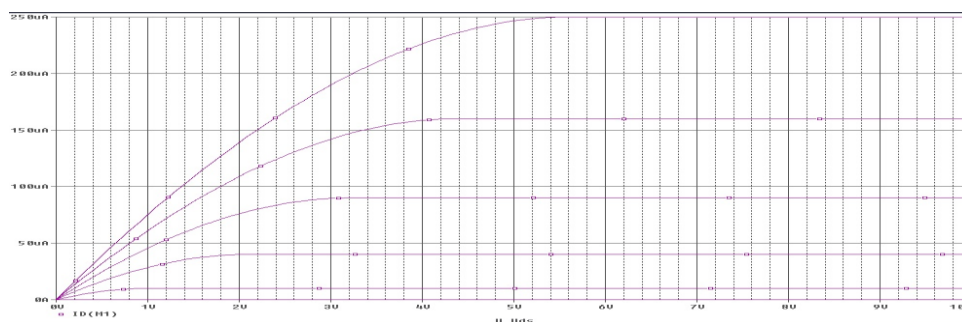
19. Observe and save the output waveform.

WAVEFORMS:

Transfer Characteristics



Drain Characteristics

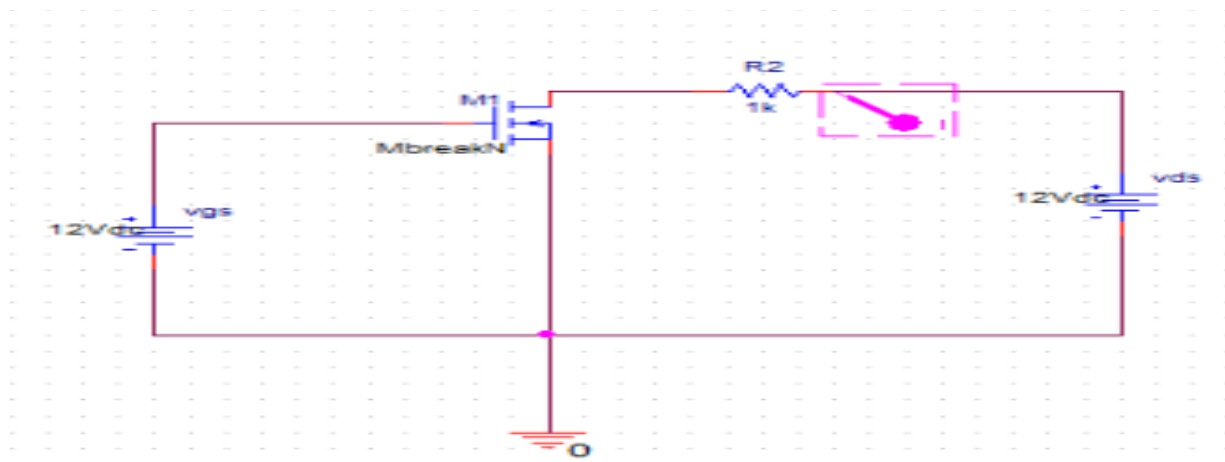


RESULT: The Transfer and Drain Characteristics of MOSFET were plotted.

EXPERIMENT No: 5

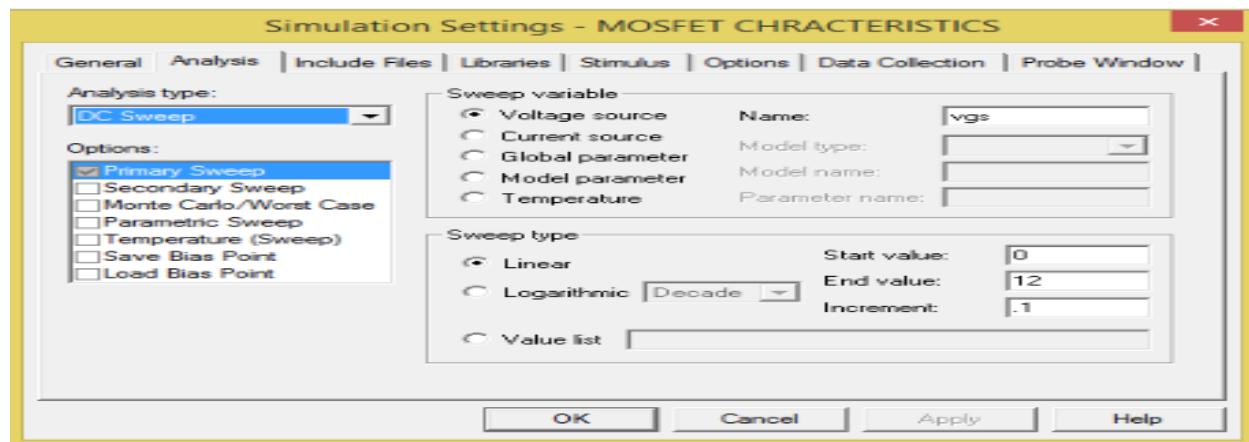
AIM : Experiments to realize Transfer and drain characteristics of a MOSFET.

MOSFET Circuit Diagram :



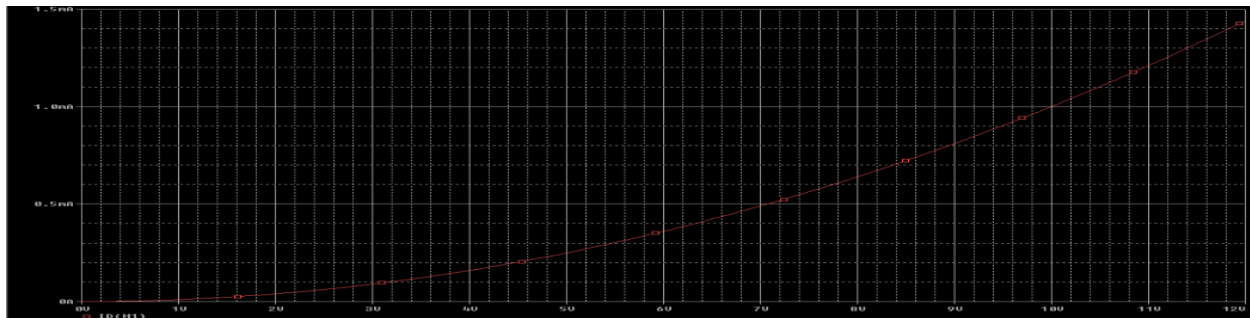
SIMULATION PROFILE SETTING :

Transfer Characteristics :



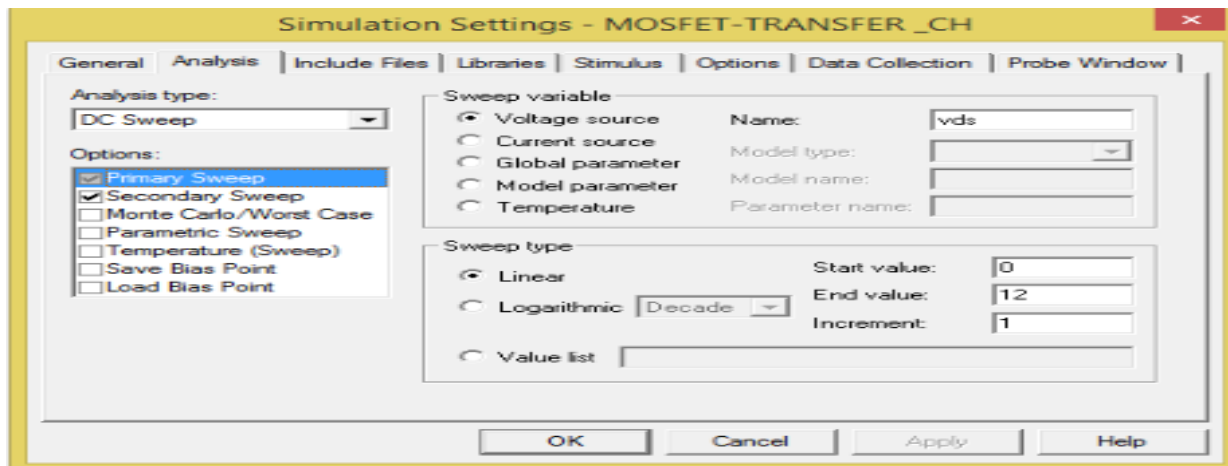
OUTPUT WAVEFORM :

Transfer Characteristics :



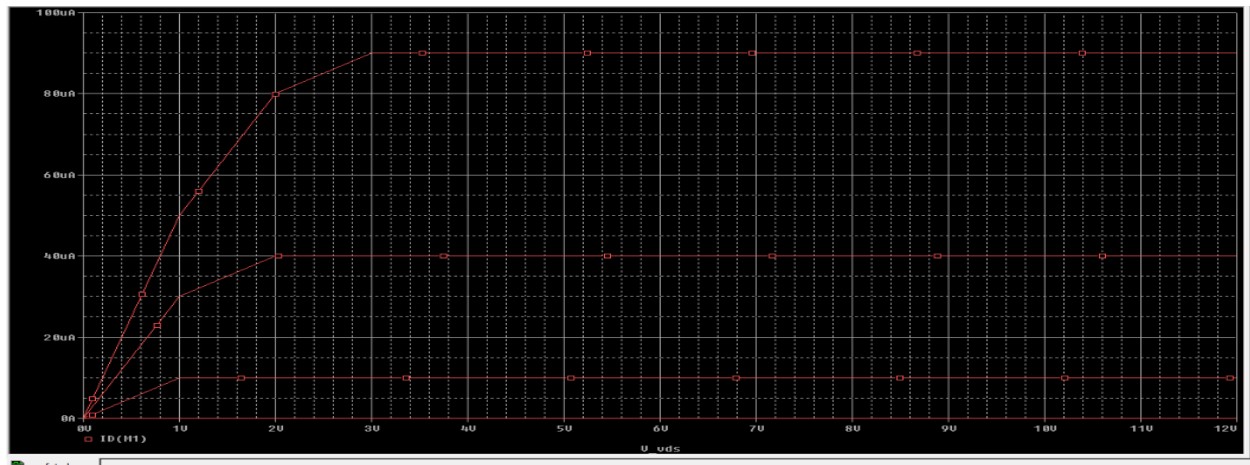
Drain Characteristics :

SIMULATION PROFILE SETTING :



OUTPUT WAVEFORM :

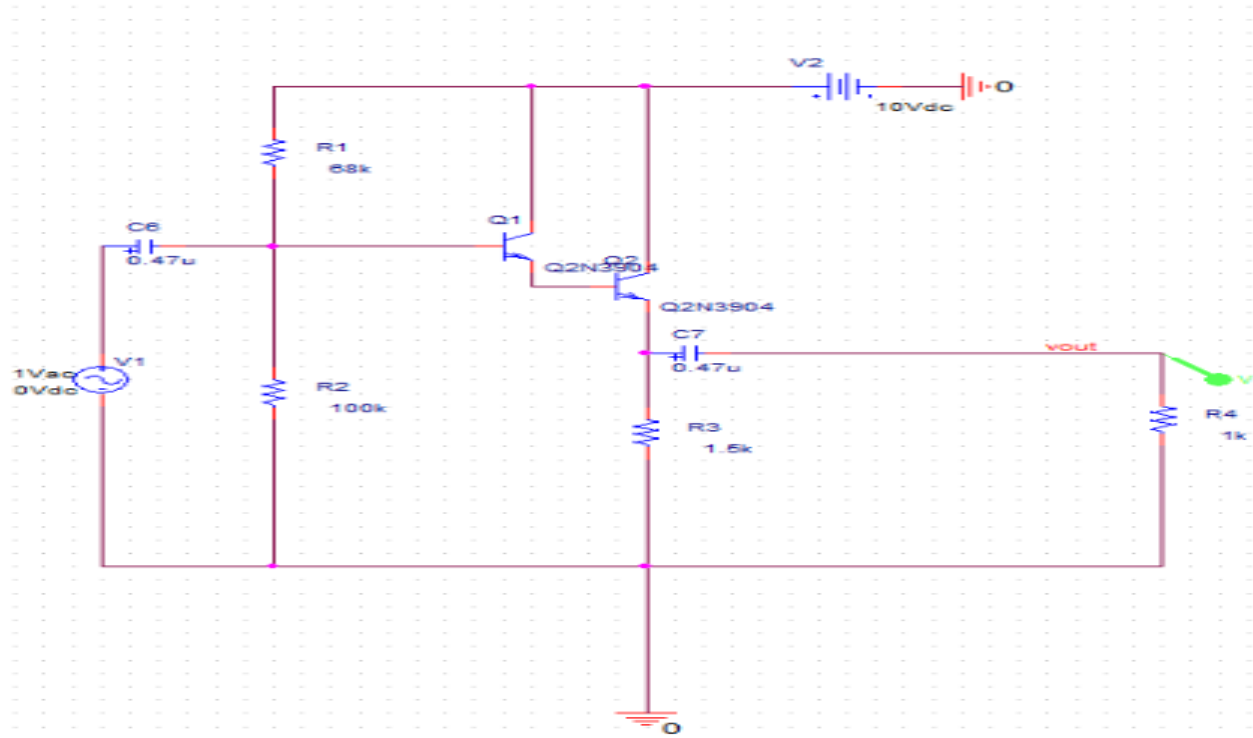
Drain Characteristics :



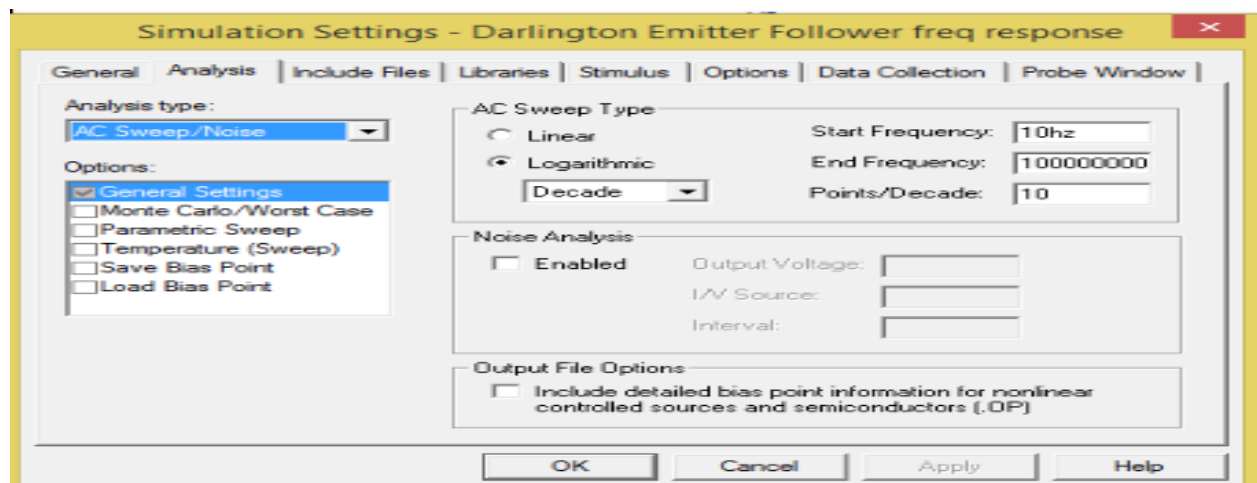
EXPERIMENT No: 6

AIM: Realize BJT Darlington Emitter follower

BJT Darlington Emitter follower Circuit Diagram :

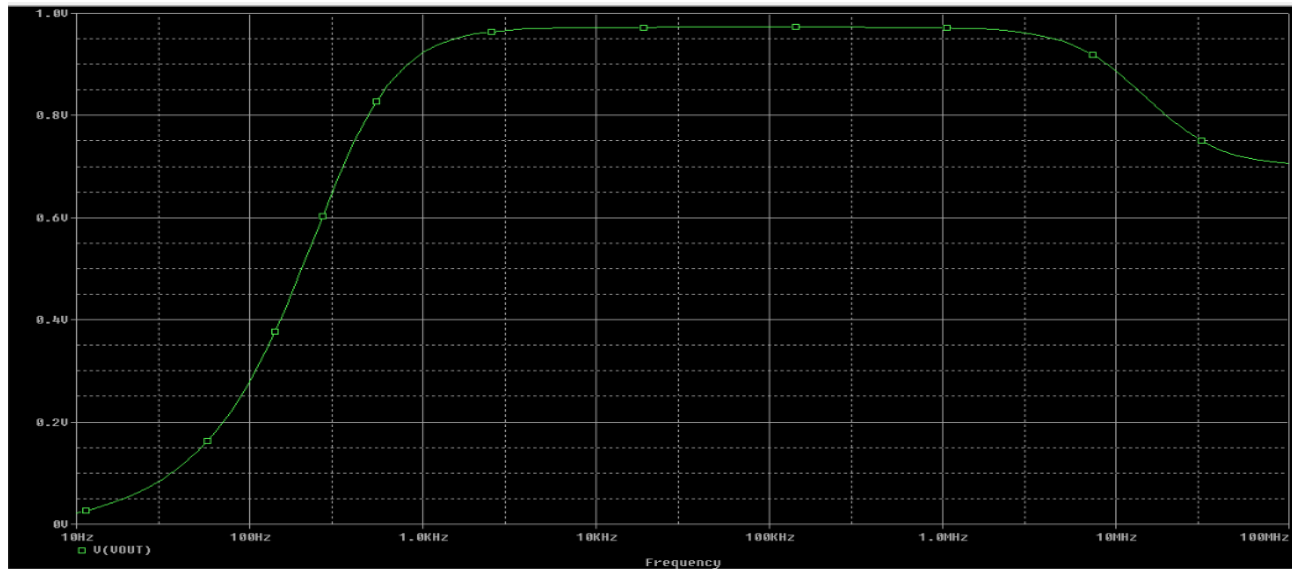


SIMULATION PROFILE SETTING :



Frequency Response :

OUTPUT WAVEFORM :

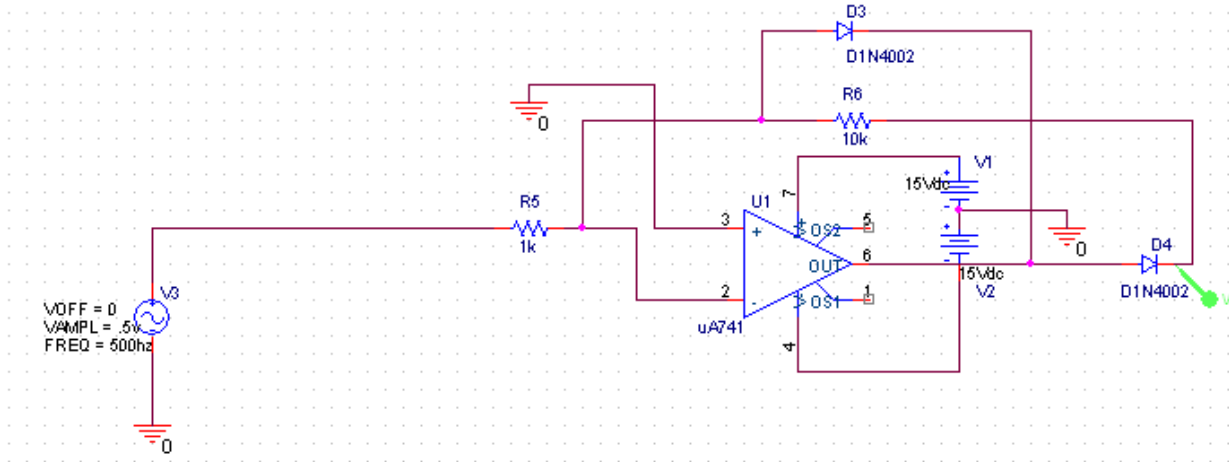


EXPERIMENT No: 9 A

PRECISION HALF WAVE RECTIFIER

AIM: To design and plot the output waveform of precision half wave rectifier using Pspice

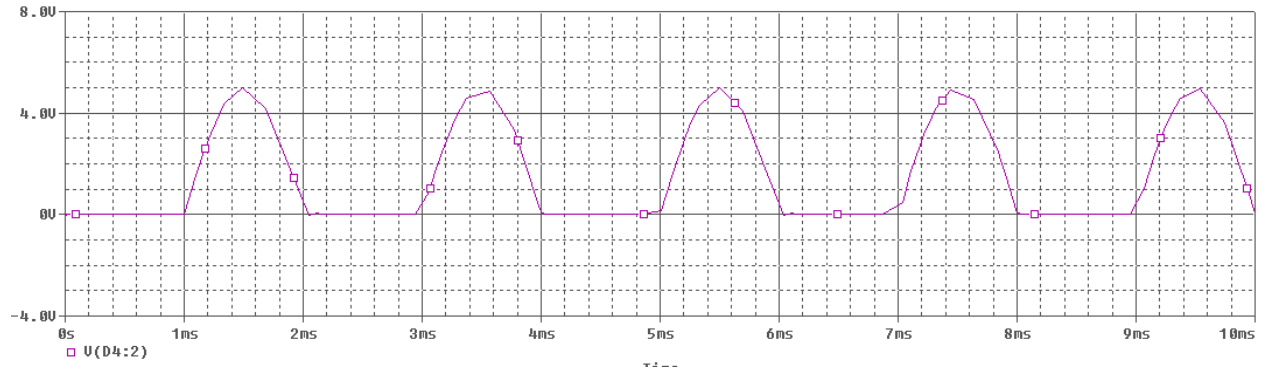
CIRCUIT DIAGRAM:



PROCEDURE:

1. Go to START → PROGRAMS→PSPICE STUDENT→CAPTURE.
2. Go to FILE→NEW→PROJECT.
3. Give a name to the Project and also select “ ANALOG AND A/D”.
4. Select the path C: ORCADDEMO/PSPICE under Browse
5. Draw the circuit as shown in the circuit diagram by using the Drag and Drop Method and selecting the required component.
6. Save it and Go to PSPICE→NEW SIMULATION PROFILE.
7. Give a name and click OK.
8. Go to PSPICE→EDIT SIMULATION PROFILE.
9. Go to ANALYSIS→TIME DOMAIN(TRANSIENTS)
10. General Settings →Run to Time:10ms
11. Start saving Data → 0 seconds
12. Save and Go to PSPICE→RUN.
13. The Output Window appears.
14. Go to ADDTRACE→ Select the appropriate parameters to be displayed.
15. Observe and save the output waveform.

WAVEFORM:

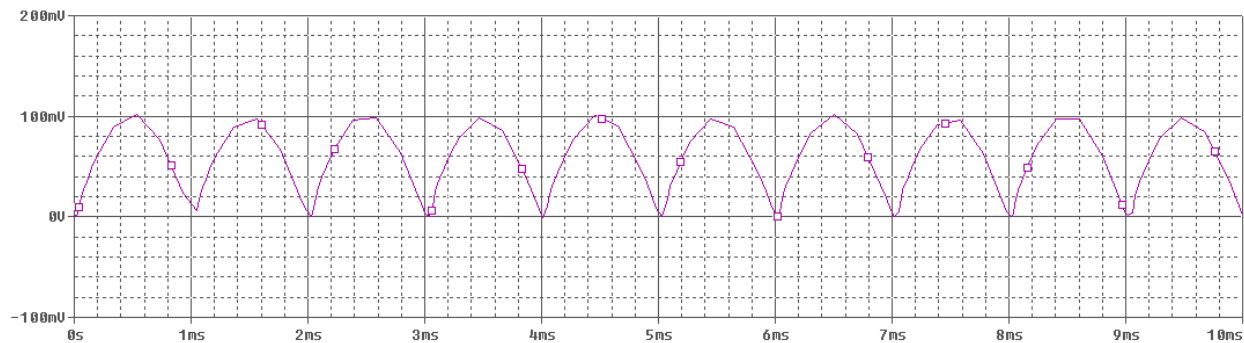


RESULTS:

The Precision Half Wave rectifier program is simulated and output waveform plotted.

10. General Settings → Run to Time: 10ms
11. Start saving Data → 0 seconds
12. Save and Go to PSPICE → RUN.
13. The Output Window appears.
14. Go to ADDTRACE → Select the appropriate parameters to be displayed.
15. Observe and save the output waveform.

WAVEFORM:



RESULTS:

The Precision Full Wave rectifier program is simulated.

EXPERIMENT No:10

RC PHASE SHIFT OSCILLATOR

AIM: To design and plot the output waveform of RC PHASE SHIFT Oscillator using Pspice and measure the frequency of oscillations

THEORY:

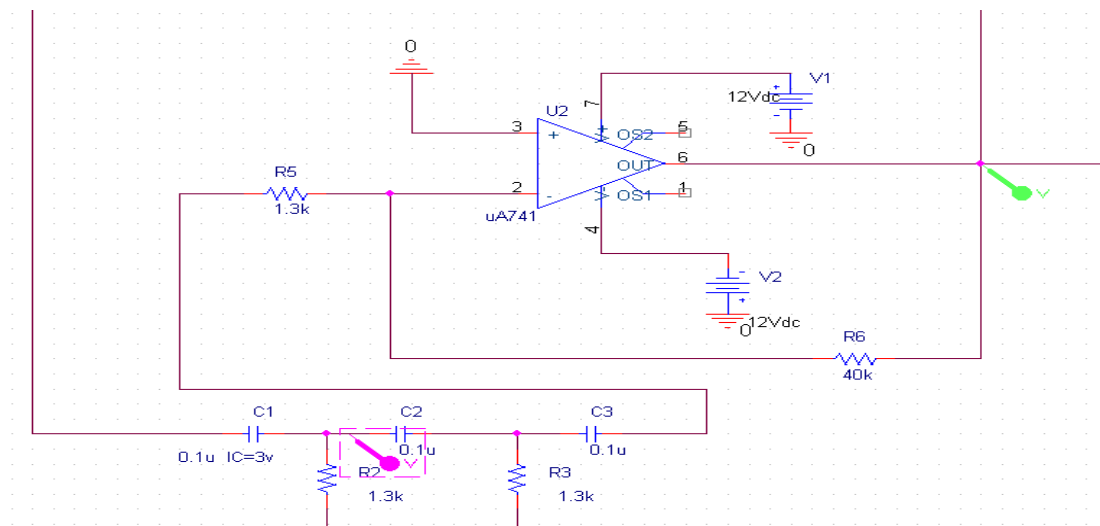
In the RC phase shift oscillator, the combination RC provides self-bias for the amplifier. The phase of the signal at the input gets reverse biased when it is amplified by the amplifier. The output of amplifier goes to a feedback network consists of three identical RC sections. Each RC section provides a phase shift of 60. Thus a total of 180. The remaining 180 phase shift is provided by the feedback network. The output of this circuit is in the same phase as the input to the amplifier.

The frequency of oscillations is given by

$$F = \frac{1}{2\pi RC} (6 + 4K)^{1/2} \text{ Where, } R_1 = R_2 = R_3 = R,$$

$$C_1 = C_2 = C_3 = C \text{ and } K = RC/R.$$

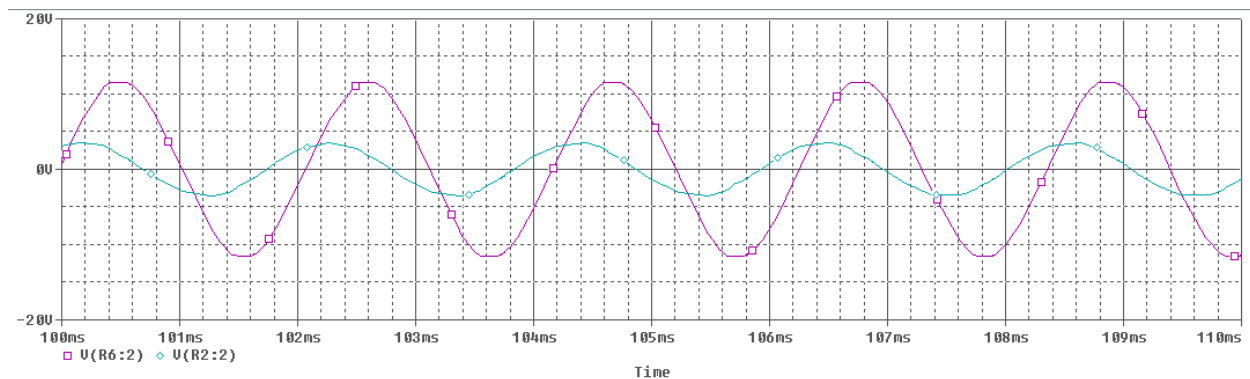
CIRCUIT DIAGRAM:



PROCEDURE:

1. Go to START → PROGRAMS→PSPICE STUDENT→CAPTURE.
2. Go to FILE→NEW→PROJECT.
3. Give a name to the Project and also select “ ANALOG AND A/D”.
4. Select the path C: ORCADDEMO/PSPICE under Browse
5. Draw the circuit as shown in the circuit diagram by using the Drag and Drop Method and selecting the required component.
6. Save it and Go to PSPICE→NEW SIMULATION PROFILE.
7. Give a name and click OK.
8. Go to PSPICE→EDIT SIMULATION PROFILE.
9. Go to ANALYSIS→TIME DOMAIN(TRANSIENTS)
10. General Settings →Run to Time:110ms
11. Start saving Data →100m seconds
12. Save and Go to PSPICE→RUN.
13. The Output Window appears.
14. Go to ADDTRACE→ Select the appropriate parameters to be displayed.
15. Observe and save the output waveform.

WAVEFORM:



RESULTS:

The RC phase shift Oscillator program is simulated and output waveform plotted.

The frequency of Oscillations: Theoretical Frequency:_____

Practical Frequency:_____